MCQ solutions

1. **C.** The quotient tends to 0, so the sum tends to 3. Neither the numerator nor the first term gives the limit.
2. **A.** Divide numerator and denominator by n^2. The dominant terms are of the same degree.
3. **B.** The product tends to $2 \times 0 = 0$. The quotient may be undefined, or change sign; the sum tends to 2.
4. **D.** $3/(n+1) < 1/10$ is equivalent to $n > 29$. At index 29, the gap is exactly 0.1.
5. **B.** The even terms equal 1, the odd ones -1; they cannot stay in the same small band around a real number. The sequence is bounded.
6. **C.** Monotone convergence. The constant sequences equal to -1 refute positivity and the limit 5; $5 - 1/(n+1)$ disproves constancy.
7. **D.** $-1/n \leq u_n \leq 1/n$, and both bounds tend to 0; no monotonicity is required.
8. **A.** For every threshold, a term exceeds it, as do all later terms since the sequence is increasing. This is the definition of the limit $+\infty$.
9. **C.** Using the conjugate, the expression is $1/(\sqrt{1+1/n}+1)$ for $n > 0$, hence $1/2$. Subtraction of infinities did not allow a conclusion.
10. **B.** The upper bound tends to 0. A sequence between the bounds can occasionally increase, for example by cancelling the even terms.

True/false. False: $(-1)^n/(n+1) \to 0$ is not monotonic. True by monotone convergence. False: $5 - 1/(n+1) \to 5$; only the weak inequality passes to the limit.

Solutions to the applications and guidance

Application 1. The difference from 5 is $2/(n+1)$; we need $n+1>40$, so $n \geq 40$. At index 39, $v_{39}=4{,}95$ is excluded. The gaps decrease, hence validity for all subsequent indices.

Application 2. The sequence has terms equal to 0 in every band, but for $\varepsilon = 1/2$, every index is followed by an odd index whose term is 1. No N guarantees the universal condition.

Application 3. The first quotient tends to 0 on dividing by n^2. The second expression is $\frac{n}{\sqrt{n^2+n}+n} = \frac{1}{\sqrt{1+1/n}+1} \to 1/2$. The third has absolute value $1/(n+1) \to 0$, so it tends to 0.

Application 4. $a_n = 2^n \to +\infty$. The equation $\ell = 2\ell$ is a necessary condition if a finite limit exists; it does not prove this existence.

Application 5. Initially $b_0 = 0$. If $0 \leq b_n < 3$, then $2 \leq b_{n+1} < 3$, so the invariant is passed on. $b_{n+1} - b_n = \frac{2}{3}(3-b_n) > 0$. Monotone convergence; the formula $b_n = 3 - 3(1/3)^n$ gives the limit 3. By continuity, $\ell = \ell/3 + 2$ also gives 3 once convergence is established.

Application 6. $(-1)^n$ is bounded between -1 and 1, not monotonic and not convergent because its even and odd subsequences have different limits. $-1/(n+1)$ is increasing, bounded between -1 and 0, and converges to 0. The sequence n is increasing, not bounded above and diverges to $+\infty$.

Key points

Have students colour the terms that stay outside the band, then express "stay" with quantifiers.
Before S32, give continuity as an accepted result in an identification exercise
or use an explicit formula. The proof of monotone convergence itself is still assumed;
students must be able to apply it reliably.

Teaching notes and solutions

Guidance for the teacher

Written test 2 comes before S19 and S20: assess the definitions and operations already met, without require monotone convergence, Bernoulli or the squeeze theorem. Draw horizontal bands around the limit and have students look for an index from which all the terms stay inside it. Require the convergence before solving the fixed-point equation. Show $(-1)^n$ to separate "bounded " and "convergent".

Solutions to the activity

1. $u_0 = 5$, $u_9 = 2.3$ and $u_{99} = 2.03$. Since $\frac{3}{n+1} > 0$ at every index, $u_n > 2$: no term equals 2.

2. $|u_n - 2| < 0{,}01 \iff \frac{3}{n+1} < 0{,}01 \iff n+1 > 300 \iff n \geq 300$. The first suitable index is so $N = 300$. At index 299, the gap is exactly 0.01 and the strict inequality is not satisfied.

3. For $\varepsilon > 0$, let us take $N = \lceil \frac{3}{\varepsilon} \rceil$, the smallest integer greater than or equal to $\frac{3}{\varepsilon}$. If $n \geq N$, then $n+1 > \frac{3}{\varepsilon}$, so $|u_n - 2| < \varepsilon$. This choice is valid, even when it is not minimal.

4. $|v_n - 2| = \frac{1}{n+1} \to 0$, therefore $v_n \to 2$. But $v_0 = 3$, $v_1 = \frac{3}{2}$ & $v_2 = \frac{7}{3}$: we observe a decrease and then a increase. This convergent sequence is not monotonic.

E1. For $n \geq 1$, $\frac{4n-1}{n+2} = \frac{4-\frac{1}{n}}{1+\frac{2}{n}} \to 4$, since the denominator tends to 1 $\neq 0$. Similarly $n^2 - 7n = n^2(1 - \frac{7}{n})$; the first factor tends to $+\infty$ and the second to 1 > 0, so the product tends to $+\infty$.

E2. The conjugate gives $\sqrt{n+1} - \sqrt{n} = \frac{(n+1)-n}{\sqrt{n+1}+\sqrt{n}} = \frac{1}{\sqrt{n+1}+\sqrt{n}}$. The denominator tends to $+\infty$, therefore the difference tends to 0.

E3. For $n \geq 1$, $-1 \leq \cos n \leq 1$ and $n > 0$, therefore $-\frac{1}{n} \leq u_n \leq \frac{1}{n}$. Both bounds tend to 0; the squeeze theorem gives $u_n \to 0$.

E4. The sequence is increasing and bounded above by 5, so it converges. Taking the limit in $u_n \leq 5$, its limit satisfies $\ell \leq 5$. It is not necessarily equal to 5: $u_n = 2 - \frac{1}{n+1}$ is increasing, bounded above by 5 and converges to 2.

E5. Since $|-0{,}6| < 1$, $(-0{,}6)^n \to 0$, therefore $u_n \to 3$. However $u_0 = 1$, $u_1 = 4{,}2$ and $u_2 = 2{,}28$: the sequence goes up then down and is not monotonic.

Solutions to homework DM2

1. Let $P(n)$: "u_n is defined and $0 \leq u_n \leq 2$". It is true at index 0 since $u_0 = 0$. Fix $n \geq 0$ and suppose $P(n)$ true. We have $2 \leq 2 + u_n \leq 4$, so $u_{n+1} = \sqrt{2+u_n}$ is defined and $\sqrt{2} \leq u_{n+1} \leq 2$. Thus $P(n+1)$ is true. By induction, all the terms belong to $[0,2]$.

2. Let $R(n)$: "$u_{n+1} \geq u_n$ ". For $n = 0$, $u_1 = \sqrt{2} \geq 0 = u_0$. Fix $n \geq 0$ and suppose $R(n)$ true. The increasing nature of $x \mapsto \sqrt{2+x}$ on $[0,2]$ gives

$$u_{n+2} = \sqrt{2+u_{n+1}} \geq \sqrt{2+u_n} = u_{n+1}.$$

   Hence $R(n+1)$ holds. By induction, (u_n) is increasing. It is bounded above by 2, so it converges to a real number $\ell \in [0,2]$.

3. The function $x \mapsto \sqrt{2+x}$ is continuous at ℓ. Taking the limit, $\ell = \sqrt{2+\ell}$. Since $\ell \geq 0$, this is equivalent to $\ell^2 - \ell - 2 = 0$, i.e. $(\ell - 2)(\ell + 1) = 0$. The root -1 is excluded; therefore $\ell = 2$.

4. Using the conjugate and $u_n \leq 2$,

$$0 \leq 2 - u_{n+1} = \frac{2 - u_n}{2 + \sqrt{2 + u_n}} \leq \frac{2 - u_n}{2}.$$

Let $T(n)$: "$0 \leq 2 - u_n \leq \frac{2}{2^n}$". At index 0, $2 - u_0 = 2$. If $T(n)$ is true for a fixed integer $n \geq 0$, then

$$0 \leq 2 - u_{n+1} \leq \frac{1}{2}\frac{2}{2^n} = \frac{2}{2^{n+1}}.$$

The base case and the inductive step prove $T(n)$ for every index. Since $\frac{2}{2^{11}} = \frac{1}{1024} < 10^{-3}$, $n = 11$ suffices. This is an index guaranteed by an upper bound, with no claim of minimality. This question may be dealt with before question 3: the number 2 already appears in the bounds.

Solutions to the textbook or past-paper work

Exercise 17. The quotients are initially of indeterminate form. For $n \geq 1$,

$$u_n = \frac{2n - 4}{n^2 + 1} = \frac{2\left(1 - \frac{2}{n}\right)}{n\left(1 + \frac{1}{n^2}\right)} \longrightarrow 0.$$

Similarly,

$$v_n = \frac{2 - 5n}{4n + 7} = \frac{\frac{2}{n} - 5}{4 + \frac{7}{n}} \longrightarrow -\frac{5}{4}, \qquad w_n = n\frac{-1 - \frac{10}{n^2} + \frac{4}{n^3}}{2 + \frac{3}{n} + \frac{1}{n^2}} \longrightarrow -\infty.$$

The last quotient tends to $-\frac{1}{2}$, and the factor n to $+\infty$.

Exercise 24. For every $n \geq 1$, $-1 \leq (-1)^n \leq 1$. Since $n^2 > 0$, we get $-\frac{1}{n^2} \leq u_n \leq \frac{1}{n^2}$. Both bounds tend to 0; the squeeze theorem gives $u_n \to 0$.